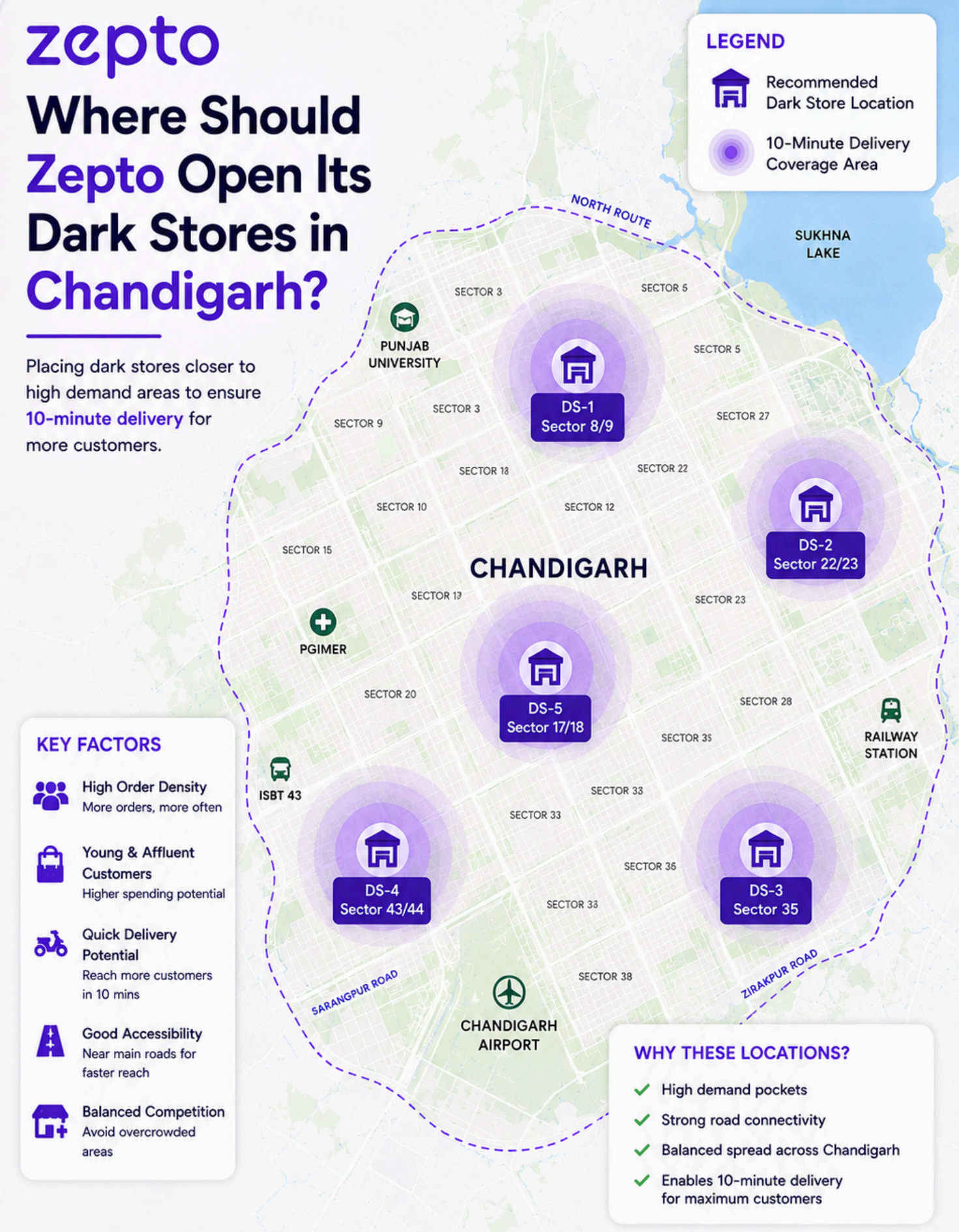




APPLIED OPERATIONS MANAGEMENT · FINAL PROJECT

INSTRUCTOR: BISMARCK

Where Should Zepto Open Its Dark Stores in Chandigarh?

A facility-location model and what happens when finance cuts the budget

Baseline → Managerial Challenge → Modified Model → New Solution → Comparison → Recommendation

Group no 8:

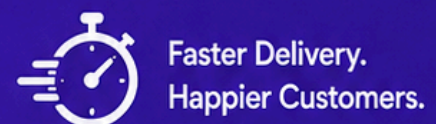
Shubham Kumar

Maneesh Baghel

Sanchit Gupta

Chandrakant

Tadi Eswar Reddy



zepto

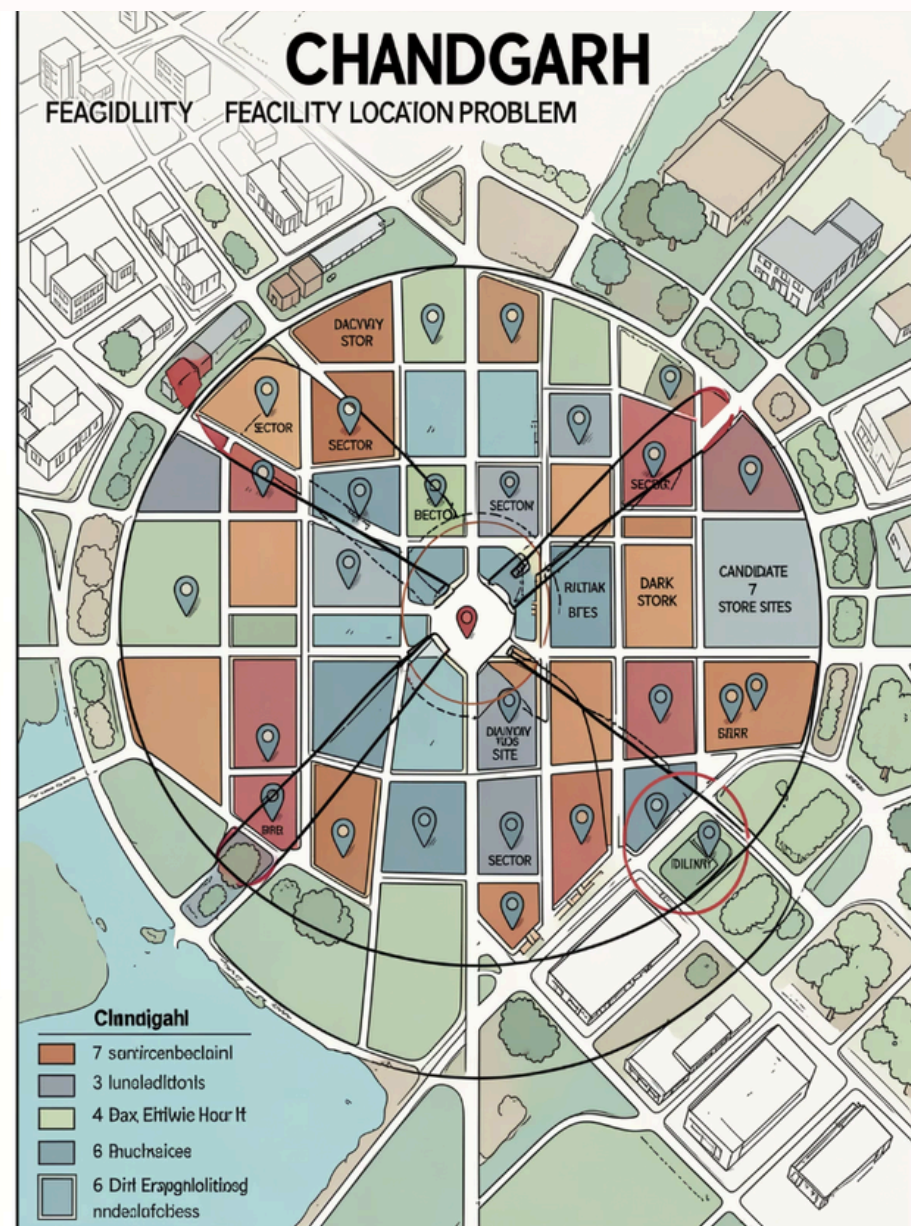


10 Minutes.
That's Zepto.

Serve All of Chandigarh Fast — Without a Store on Every Corner

Zepto is entering Chandigarh with a **~10-minute delivery promise**. A dark store must sit near every customer — but each carries a heavy fixed cost. We must choose **which sites to open** and **which sectors each serves** — balancing cost, service, and accessibility across **16 sectors** and **7 candidate sites**.

Scope: 16 sectors · 7 sites · ~63,100 orders/month · real GPS distances · delivery ₹10/order-km



The Integer Programme

Objective:

$$\min \sum_j f_j y_j + c \cdot \sum_{i,j} P_i d_{ij} x_{ij}$$

Store fixed cost + delivery cost

Subject to:

$$\sum_j x_{ij} = 1 \quad \forall i$$

Each sector served exactly once

$$x_{ij} \leq y_j \quad \forall i, j$$

Serve only from open stores

$$\sum_i P_i x_{ij} \leq 0.75 C_j y_j \quad \forall j$$

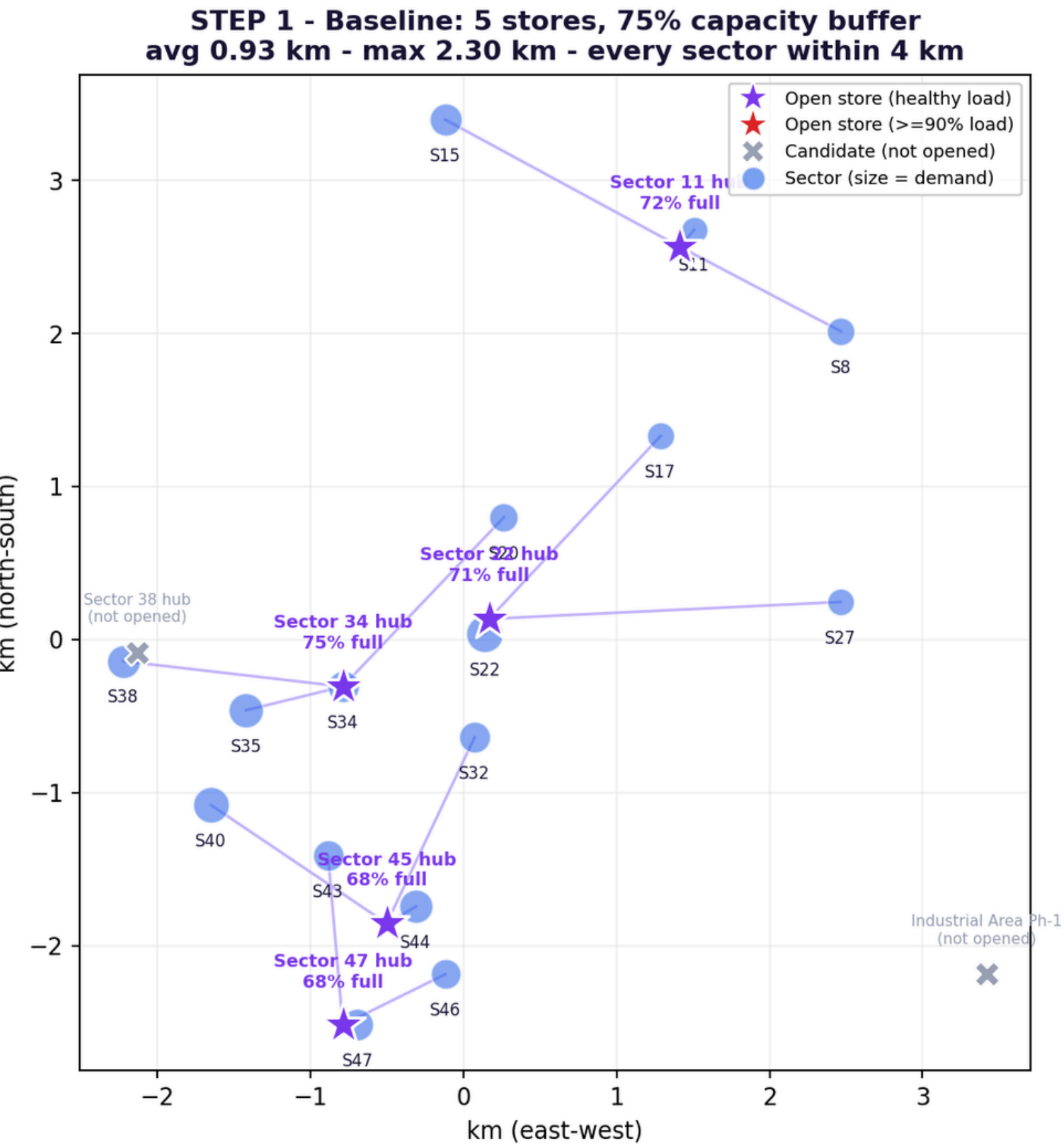
Capacity with 75% safety buffer

$$d_{ij} > 4 \text{ km} \implies x_{ij} = 0$$

10-minute delivery radius

Decisions: $y_j = 1$ if store j opens; $x_{ij} = 1$ if sector i served from j

The Optimal Network: Open 5 of 7 Stores



Total cost = ₹12.9 L store rent + ₹5.9 L delivery. Every sector is within the 10-minute promise and no store exceeds 75% load — slack to absorb demand spikes.

5/7

Stores Opened

₹18.8L

Total Cost / Month

₹12.9L fixed + ₹5.9L delivery

2.30km

Worst-Case Distance

0.93km

Avg Distance
Demand-weighted

68-75%

Store Load

Healthy slack maintained

Then the Manager Changes the Rules

"Finance approved capex for only 4 stores, not 5. Get me the best plan you can — but four is the hard limit."

A Brand-New Restriction

The baseline opened 5 stores. A cap of 4 is a binding constraint the original model never had.

Same Demand, Fewer Stores

Fewer, larger stores must absorb the same **631 units of demand** — straining capacity.

The Core Question

Can we even serve Chandigarh with 4 stores — and at what cost to speed and resilience?

One New Constraint — That Breaks Feasibility

Add the Budget Constraint

$$\sum_j y_j \leq 4$$

Solver says: **INFEASIBLE**

The 4 largest stores hold only **578 units** at the 75% buffer — but demand is **631**. The promise can't be kept.

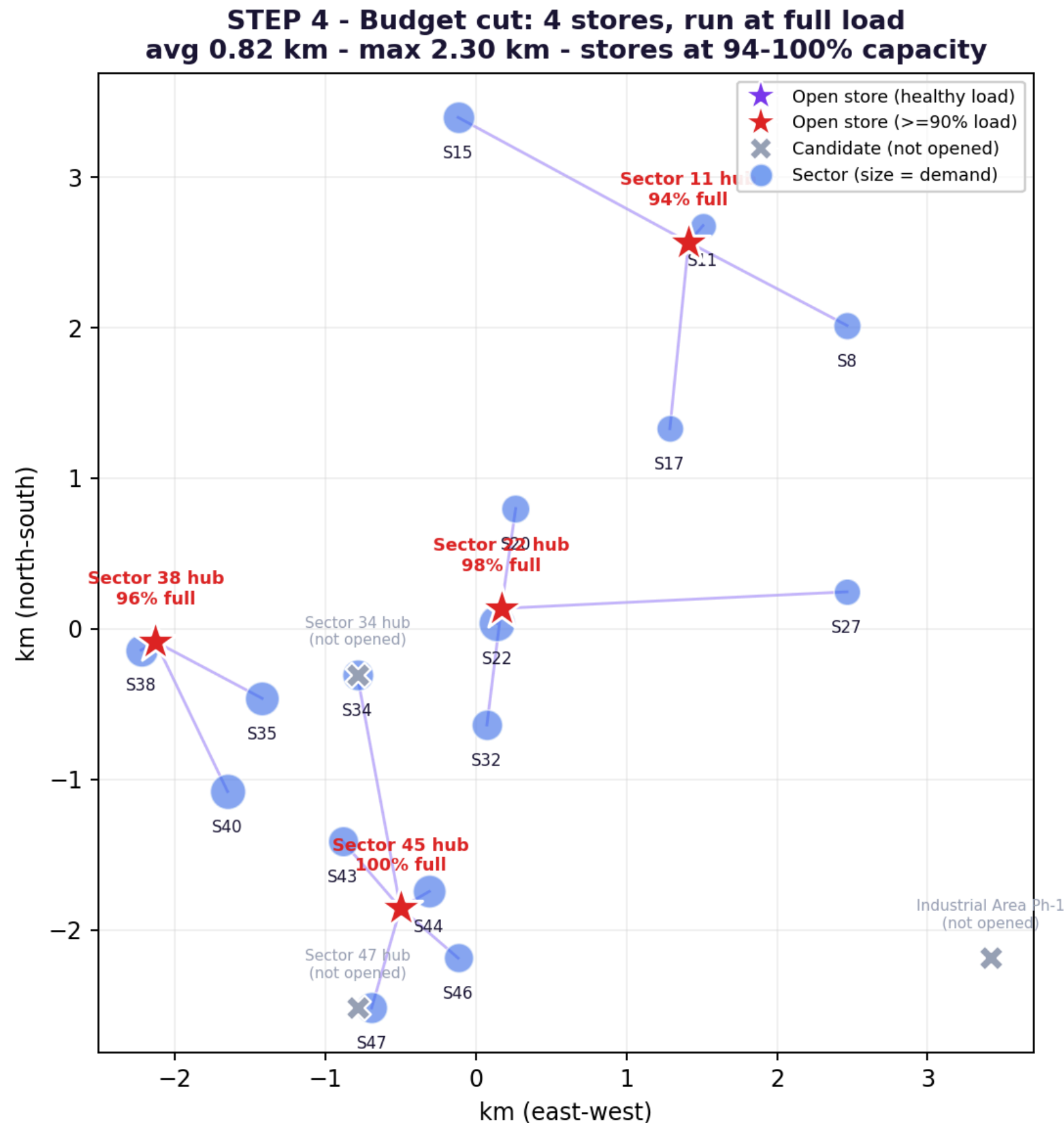
The Managerial Fix: Relax the Safety Buffer

To stay within 4 stores we must let them run at full capacity, the 0.75 buffer becomes 1.0. The budget cut forces **two** model changes, not one:

$$\sum_i P_{ij} x_{ij} \leq 0.75 C_j y_j \quad \rightarrow \quad \sum_i P_{ij} x_{ij} \leq 1.00 C_j y_j$$

Trade-off in plain terms: To meet the budget, every store runs at full load — we trade financial cost for operational fragility.

4-Store Network: Cheaper, Same Distance, But Maxed Out



Cheaper by ₹3.8 L/month and distance is actually slightly better but every store now runs at **94–100% capacity**. One store is at its absolute limit.

4/7
Stores Opened

₹15.0L
Total Cost / Month
–₹3.8L vs. baseline

2.30km
Worst-Case Distance
Unchanged

0.82km
Avg Distance
Slightly better!

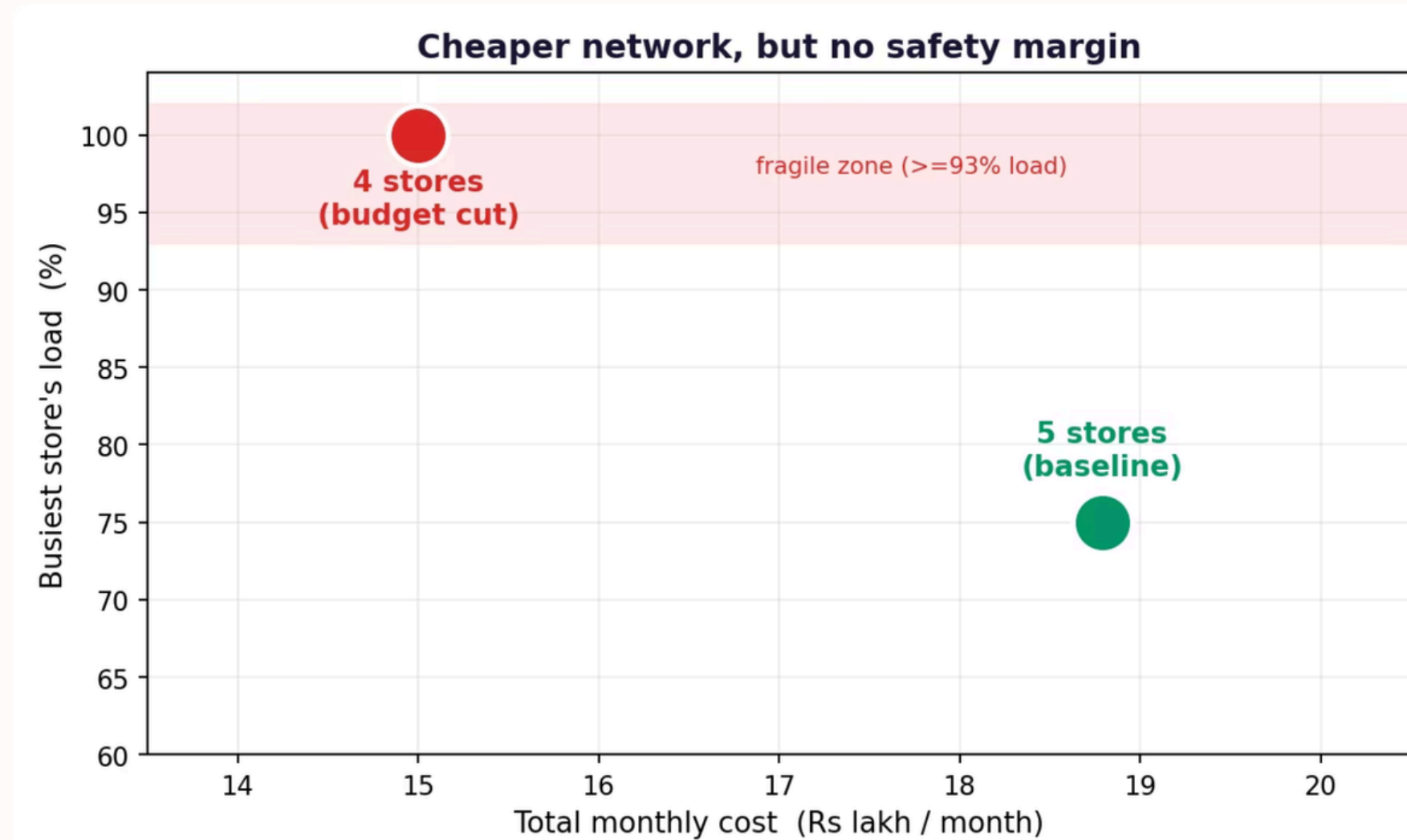
94–100%
Store Load
No slack at all

Baseline vs. Budget-Cut, Side by Side

Metric	Baseline (5 stores)	Budget Cut (4 stores)
Stores Opened	5 of 7	4 of 7
Store (Fixed) Cost	₹12.9 lakh	₹9.8 lakh
Delivery Cost	₹5.9 lakh	₹5.2 lakh
Total Cost / Month	₹18.8 lakh	₹15.0 lakh (−20%)
Avg Distance	0.93 km	0.82 km (better)
Worst-Case Distance	2.30 km	2.30 km (same)
Busiest Store Load	75%	100% (no slack)
Capacity Safety Buffer	Yes (25%)	None — full load

Same model, one extra constraint — the table makes the cost of the budget cut fully transparent to the manager.

Our Recommendation to the Manager



What We Advise

- If 4 stores is a hard cap, it is workable — open **W3, W4, W5, W6**.
- Flag the risk: stores at 94–100% load have zero headroom for demand spikes quick-commerce sees daily (rain, weekends, festivals).
- Treat 4 stores as a **launch configuration**; phase in the 5th store as revenue grows to restore the safety buffer.
- The catch: the cut barely changes delivery distance — its real cost is fragility, not service. The extra ₹3.8 L/month buys resilience.

Financial Impact

₹3.8 L/month saving ($\approx$ ₹45 L/year) — real, but buys fragility, not distance.

Operational Risk

Zero headroom for demand spikes — rain, weekends, festivals.

Suggested Rollout

Launch with 4 stores; phase in the 5th as revenue grows to restore the safety buffer.

👉 **Data → model → constraint → re-solve → decision:** The budget question is now answered with evidence, not intuition.